

Tabular Planning: Policy Iteration

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CS 1840 Fall 2026

Announcements

HW1 is out!

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Recap: Bellman Optimality

$$\mathcal{M} = \{S, A, P, r, \gamma\}$$

$$P : S \times A \mapsto \Delta(S), \quad r : S \times A \rightarrow [0,1], \quad \gamma \in [0,1)$$

$$\text{Policy } \pi : S \mapsto A$$



Recap: Bellman Optimality

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Bellman Optimality—the Q version

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a' \in A} Q^*(s', a') \right]$$

(Note: Red handwritten marks are present in the original image: a triangle under $Q^(s, a)$, a triangle under $r(s, a)$, a circle around γ , and a triangle under $\max_{a' \in A}$)*

Finishing Up From Last Lecture



1: Bellman Optimality

2: An Iterative Algorithm: Value Iteration

Value Iteration Algorithm:

1. Initialization: $Q^0 \in [0, \frac{1}{1-\gamma}]$

2. Iterate until convergence: $Q^{t+1} \leftarrow \mathcal{T} Q^t$

Value Iteration Algorithm:

1. Initialization: Q^0

2. Iterate until convergence: $Q^{t+1} \Leftarrow \mathcal{T} Q^t$

We hope $Q^t \rightarrow Q^\star$, as $t \rightarrow \infty$
 Δ

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_{\infty} \leq \gamma \|Q - Q'\|_{\infty}$$

Proof:

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty$$

Proof:

$$|(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| = \left| \underbrace{r(s, a)}_{\text{deterministic}} + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(\underbrace{r(s, a)}_{\text{deterministic}} + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right|$$

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Proof:

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$$| \mathbb{E}_{x \sim p} f(x) - \mathbb{E}_{x \sim p} g(x) |$$

$$\leq \mathbb{E}_{x \sim p} |f(x) - g(x)|$$

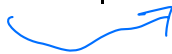
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$$\begin{aligned} &| \max_a f(a) - \max_a g(a) | \\ &\leq \max_a |f(a) - g(a)| \end{aligned}$$

Convergence of Value Iteration:

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Convergence of Value Iteration:

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$Q \in \mathbb{R}^{S \times A}$

$$\|x\|_\infty = \max_{1 \leq i \leq d} |x_i|$$

Convergence of Value Iteration:

Lemma [Convergence]: Given Q^0 , we have:

$$\|Q^t - Q^\star\|_\infty \leq \gamma^t \|Q^0 - Q^\star\|_\infty$$

Convergence of Value Iteration:

Lemma [Convergence]: Given Q^0 , we have:

$$\|Q^t - Q^*\|_\infty \leq \gamma^t \|Q^0 - Q^*\|_\infty$$

by definition
↓

$$\|Q^{t+1} - Q^*\|_\infty = \|\mathcal{T}Q^t - \mathcal{T}Q^*\|_\infty \leq \gamma \|Q^t - Q^*\|_\infty$$

Convergence of Value Iteration:

Lemma [Convergence]: Given Q^0 , we have:

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$$\|Q^{t+1} - Q^\star\|_\infty = \|\mathcal{T}Q^t - \mathcal{T}Q^\star\|_\infty \leq \gamma \|Q^t - Q^\star\|_\infty$$

$$\dots \leq \gamma^{t+1} \|\widehat{Q}^0 - Q^\star\|_\infty$$

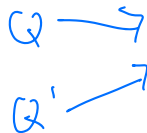
Value Iteration

1. VI

(a fix point iteration alg):

$$Q^{t+1} \leftarrow \mathcal{T} Q^t$$

↑ optimality
Bellman
operator



Value Iteration

1. VI

(a fix point iteration alg):

$$Q^{t+1} \Leftarrow \mathcal{T} Q^t$$

Contraction



2. VI convergence: exponentially fast,
i.e., $\|Q^t - Q^*\|_\infty \leq \gamma^t \|Q^0 - Q^*\|_\infty$

Value Iteration

1. VI
(a fix point iteration alg): $Q^{t+1} \leftarrow \mathcal{T} Q^t$ Contraction $\longrightarrow$ 2. VI convergence: exponentially fast,
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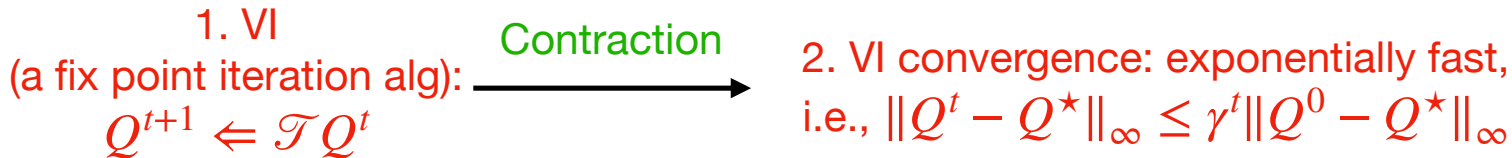
1. How to extract a policy from VI?

$$Q^t$$
$$\pi^t(s) = \underset{a}{\operatorname{argmax}} Q^t(s, a)$$

$$Q^t \approx Q^* \quad t \rightarrow \infty$$

$$\pi^*(s) = \underset{a}{\operatorname{argmax}} Q^*(s, a)$$

Value Iteration



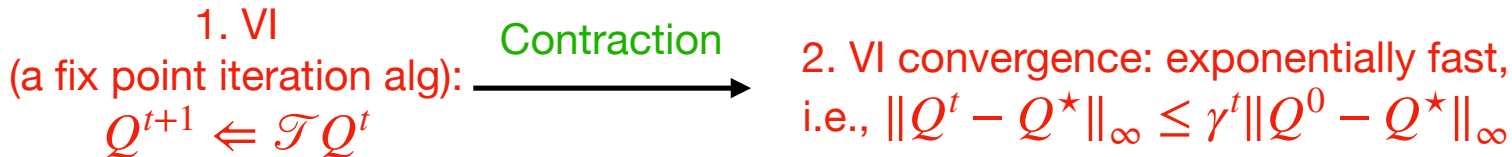
1. How to extract a policy from VI?

2. We could set $\pi^t(s) = \arg \max_a Q^t(s, a)$, does $\pi^t \rightarrow \pi^*$ when t increases?

$$\begin{aligned}
 Q^*(s, a_1) &= 100 \\
 Q^*(s, a_2) &= 100 + 1e^{-5} \\
 Q^*(s, a_1) - Q^*(s, a_2) &= 1e^{-5}
 \end{aligned}$$

$$\begin{aligned}
 t &= 1000 \\
 Q^t(s, a_1) &= 100 + 1e^{-4} \\
 Q^t(s, a_2) &= 100 + 1e^{-5} \\
 \|Q^t(s, a_1) - Q^t(s, a_2)\|_\infty &= 1e^{-4}
 \end{aligned}$$

Value Iteration



1. How to extract a policy from VI?

2. We could set $\pi^t(s) = \arg \max_a Q^t(s, a)$, does $\pi^t \rightarrow \pi^*$ when t increases?

3. Can we still hope π^t being a good policy?

Value Iteration

Value Iteration

Policy Performance: $\underline{V^{\pi^t}(s)} \geq \underline{V^*(s)} - \frac{2}{1-\gamma} \|Q^t - Q^*\|_\infty \forall s \in S$

$$V^{\pi^t}(s) - V^*(s) \geq - \frac{2}{1-\gamma} \|Q^t - Q^*\|_\infty \quad \forall s \in S$$

Value Iteration

Policy Performance: $V^{\pi^t}(s) \geq V^*(s) - \frac{2}{1-\gamma} \|Q^t - Q^*\|_\infty \forall s \in S$

Error in Q is amplified by $1/(1 - \gamma)$

Value Iteration

Policy Performance: $V^{\pi^t}(s) \geq V^*(s) - \frac{2}{1-\gamma} \|Q^t - Q^*\|_\infty \forall s \in S$

Error in Q is amplified by $1/(1 - \gamma)$

(Because π^t could disagree w/ π^* at every step)

Question for Today:

Given an MDP $\mathcal{M} = (S, A, P, r, \gamma)$, How to directly search for $\pi^\star : S \mapsto A$

Outline:

1: An Iterative Algorithm: Policy Iteration

2: Convergence? How fast?

3: A new model: Finite horizon MDP

Algorithm: Policy Iteration

1. Initialization: $\pi^0 : S \mapsto \Delta(A)$

2. For $t = 0 \dots$,

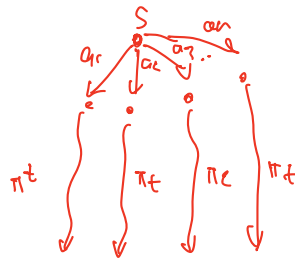
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Algorithm: Policy Iteration

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2. For $t = 0 \dots$,

Policy Evaluation: compute $Q^{\pi^t}(s, a), \forall s, a$



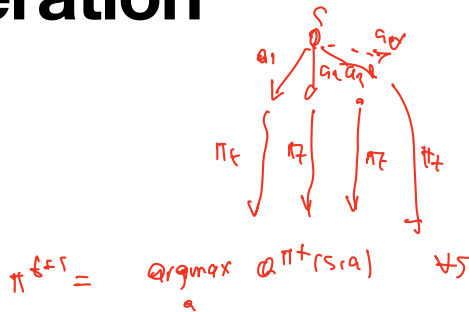
Algorithm: Policy Iteration

1. Initialization: $\pi^0 : S \mapsto \Delta(A)$

2. For $t = 0 \dots$,

Policy Evaluation: compute $Q^{\pi^t}(s, a), \forall s, a$

Policy Improvement $\pi^{t+1}(s) := \arg \max_a Q^{\pi^t}(s, a), \forall s$



Outline:



1: An Iterative Algorithm: Policy Iteration

2: Convergence? How fast?

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Key properties of Policy Iterations:

① Monotonic improvement:

$$Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$$

$$Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a) \quad \forall s, a$$

$$\arg \max_a Q^{\pi^t}(s, a) \subseteq \arg \max_a Q^{\pi^{t+1}}(s, a)$$

2. When $\pi^{t+1} = \pi^t$, then π^t is equal to π^*

Monotonic Improvement

Recall: Policy Improvement $\pi^{t+1}(s) = \arg \max_a Q^{\pi^t}(s, a), \forall s$

Monotonic improvement $Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$

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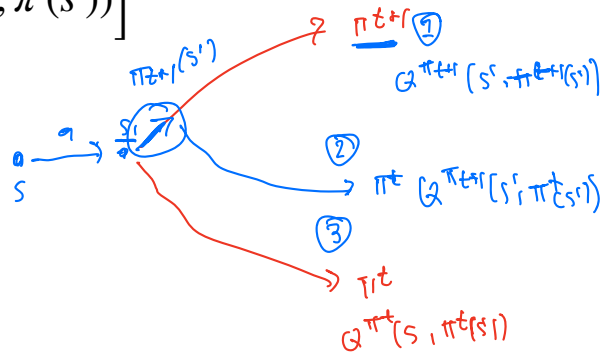
Monotonic improvement $Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$

$$\overset{\textcircled{1}}{Q^{\pi^{t+1}}(s, a)} - \overset{\textcircled{2}}{Q^{\pi^t}(s, a)} = \gamma \mathbb{E}_{s' \sim P(s, a)} \left[\overset{\textcircled{3}}{Q^{\pi^{t+1}}(s', \pi^{t+1}(s'))} - Q^{\pi^t}(s', \pi^t(s')) \right]$$

$$Q^{\pi^{t+1}}(s, a) = \cancel{r(s, a)} + \gamma \mathbb{E}_{s' \sim P} [Q^{\pi^{t+1}}(s', \pi^{t+1}(s'))]$$

$$Q^{\pi^t}(s, a) = \cancel{r(s, a)} + \gamma \mathbb{E}_{s' \sim P} [Q^{\pi^t}(s', \pi^t(s'))]$$

$$\textcircled{1} - \textcircled{2} + \textcircled{2} - \textcircled{3}$$



Monotonic Improvement

Recall: Policy Improvement $\pi^{t+1}(s) = \arg \max_a Q^{\pi^t}(s, a), \forall s$

Monotonic improvement $Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$

$$\begin{aligned}
 Q^{\pi^{t+1}}(s, a) - Q^{\pi^t}(s, a) &= \gamma \mathbb{E}_{s' \sim P(s, a)} \left[Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s')) \right] \geq 0 \\
 &= \gamma \mathbb{E}_{s' \sim P(s, a)} \left[\underbrace{Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^{t+1}(s'))}_{\text{①}} + \underbrace{Q^{\pi^t}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s'))}_{\text{②}} \right]
 \end{aligned}$$

$Q^{\pi^t}(s', \pi^{t+1}(s')) = \max_a Q^{\pi^t}(s', a) \geq Q^{\pi^t}(s', \pi^t(s'))$

Monotonic Improvement

Recall: Policy Improvement $\pi^{t+1}(s) = \arg \max_a Q^{\pi^t}(s, a), \forall s$

Monotonic improvement $Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$

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$$= \gamma \mathbb{E}_{s' \sim P(s, a)} \left[Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^{t+1}(s')) + Q^{\pi^t}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s')) \right]$$

Monotonic Improvement

Recall: Policy Improvement $\pi^{t+1}(s) = \arg \max_a Q^{\pi^t}(s, a), \forall s$

Monotonic improvement $Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$

$$\begin{aligned}
 \underline{Q^{\pi^{t+1}}(s, a)} - \underline{Q^{\pi^t}(s, a)} &= \gamma \mathbb{E}_{s' \sim P(s, a)} \left[Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s')) \right] \\
 &= \gamma \mathbb{E}_{s' \sim P(s, a)} \left[Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^{t+1}(s')) + \underbrace{Q^{\pi^t}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s'))}_{\geq 0} \right] \\
 &\geq \gamma \mathbb{E}_{s' \sim P(s, a)} \left[\underline{Q^{\pi^{t+1}}(s', \pi^{t+1}(s'))} - \underline{Q^{\pi^t}(s', \pi^{t+1}(s'))} \right]
 \end{aligned}$$

Handwritten notes and diagrams:

- Red dashed oval around the term $Q^{\pi^t}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s'))$ with a circled 2 below it.
- Blue circles around s' and $\pi^{t+1}(s')$ in the last term.
- Handwritten diagram showing a state s' branching into actions $\pi^{t+1}(s')$, $Q^{\pi^{t+1}}(s', \pi^{t+1}(s'))$, and $\pi^t(s')$.

Monotonic Improvement

Recall: Policy Improvement $\pi^{t+1}(s) = \arg \max_a Q^{\pi^t}(s, a), \forall s$

Monotonic improvement $Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$

$$\begin{aligned}
 [Q^{\pi^{t+1}}(s, a) - Q^{\pi^t}(s, a)] &= \gamma \mathbb{E}_{s' \sim P(s, a)} [Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s'))] \\
 &= \gamma \mathbb{E}_{s' \sim P(s, a)} [\underbrace{Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^{t+1}(s'))}_{\text{red}}] + \underbrace{[Q^{\pi^t}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^t(s'))]}_{\text{blue}}] \\
 (\geq) \gamma \mathbb{E}_{s' \sim P(s, a)} [\underbrace{Q^{\pi^{t+1}}(s', \pi^{t+1}(s')) - Q^{\pi^t}(s', \pi^{t+1}(s'))}_{\text{blue}}] &\geq \underbrace{(\dots)}_{\text{blue}} \geq -\gamma^\infty / (1 - \gamma) = 0
 \end{aligned}$$

$\leftarrow \left[-\frac{1}{1-\gamma}, \frac{1}{1-\gamma} \right]$
 $Q^{\pi^{t+1}}(s, a) - Q^{\pi^t}(s, a) \geq 0$

Monotonic Improvement

Recall: Policy Improvement $\pi^{t+1}(s) = \arg \max_a Q^{\pi^t}(s, a), \forall s$

Monotonic improvement $Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$

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$$V^{\pi^{t+1}}(s) \geq V^{\pi^t}(s), \forall s, ??$$

$$V^{\pi^{t+1}}(s) = Q^{\pi^{t+1}}(s, \pi^{t+1}(s)) \geq Q^{\pi^t}(s, \pi^{t+1}(s)) \geq Q^{\pi^t}(s, \pi^t(s))$$

↑
causes max
a

Summary of Policy Iteration

Iterate between Policy Evaluation and Policy Improvement:

$$\pi^{t+1}(s) := \arg \max_a Q^{\pi^t}(s, a), \forall s$$

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Monotonic improvement

$$Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$$

Outline:



1: An Iterative Algorithm: Policy Iteration



2: Convergence? How fast?

3: A new model: Finite horizon MDP

Finite horizon Markov Decision Process

$$\mathcal{M} = \{S, A, r, P, H, \mu_0\},$$

Horizon (points to H)
initial state distribution (points to μ_0)

$$r : S \times A \mapsto [0, 1], H \in \mathbb{N}^+, P : S \times A \mapsto \Delta(S), s_0 \sim \mu_0$$

Finite horizon Markov Decision Process

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$$r : S \times A \mapsto [0,1], H \in \mathbb{N}^+, P : S \times A \mapsto \Delta(S), s_0 \sim \mu_0$$

i.e., the task always starts from $s_0 \sim \mu_0$, and lasts for H total steps

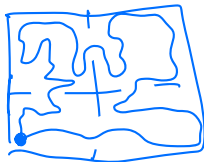
Finite horizon Markov Decision Process

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i.e., the task always starts from $s_0 \sim \mu_0$, and lasts for H total steps

Very common in control,
e.g., keep tracking a pre-specified trajectory with fixed length and fixed initial state

vacuum cleaner



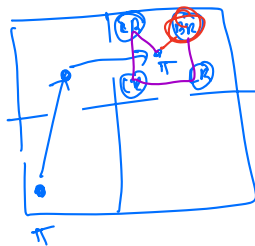
Finite horizon Markov Decision Process

$$\mathcal{M} = \{S, A, r, P, H, \mu_0\},$$

$$r : S \times A \mapsto [0,1], H \in \mathbb{N}^+, P : S \times A \mapsto \Delta(S), s_0 \sim \mu_0$$

Note that in finite horizon setting, we will consider **time-dependent policies**, i.e.,

$$\pi := \{ \underset{\Delta}{\pi_0}, \underset{\Delta}{\pi_1}, \dots, \underset{\Delta}{\pi_{H-1}} \}, \pi_h : S \mapsto A, \forall h$$



BR = Big Reward
 LR = Little Rewards

π_{H-2}

π_{H-2}

Finite horizon Markov Decision Process

$$\mathcal{M} = \{S, A, r, P, H, \mu_0\},$$
$$r : S \times A \mapsto [0,1], H \in \mathbb{N}^+, P : S \times A \mapsto \Delta(S), s_0 \sim \mu_0$$

Note that in finite horizon setting, we will consider **time-dependent policies**, i.e.,

$$\pi := \{\pi_0, \pi_1, \dots, \pi_{H-1}\}, \pi_h : S \mapsto A, \forall h$$

Policy interacts with the MDP as follows:

$$\tau = \{s_0, a_0, s_1, a_1, \dots, s_{H-1}, a_{H-1}\}, \boxed{s_0 \sim \mu_0, a_0 = \pi_0(s_0), s_1 \sim P(\cdot | s_0, a_0), a_1 = \pi_1(s_1), \dots}$$

V/Q functions in Finite horizon MDP

$$V_h^\pi(s) = \mathbb{E} \left[\sum_{t=h}^{H-1} r(s_t, a_t) \mid s_h = s, \pi \right]$$

V/Q functions in Finite horizon MDP

$$V_h^\pi(s) = \mathbb{E} \left[\sum_{\tau=h}^{H-1} r(s_\tau, a_\tau) \mid s_h = s, \pi \right]$$

$$Q_h^\pi(s, a) = \mathbb{E} \left[\sum_{\tau=h}^{H-1} r(s_\tau, a_\tau) \mid (s_h, a_h) = (s, a), a_\tau \sim \pi \text{ for } \tau > h \right]$$

Bellman Equation

$$Q_h^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim p} [V_{h+1}^\pi(s')]]$$